

Accepted Manuscript

Removal of antibiotic compounds by adsorption using glycerol-based carbon materials

S. Álvarez-Torrellas, R.S. Ribeiro, H.T. Gomes, G. Ovejero, J. García

PII: S1385-8947(16)30365-5
DOI: <http://dx.doi.org/10.1016/j.ccej.2016.03.112>
Reference: CEJ 14964

To appear in: *Chemical Engineering Journal*

Received Date: 30 January 2016
Revised Date: 21 March 2016
Accepted Date: 22 March 2016



Please cite this article as: S. Álvarez-Torrellas, R.S. Ribeiro, H.T. Gomes, G. Ovejero, J. García, Removal of antibiotic compounds by adsorption using glycerol-based carbon materials, *Chemical Engineering Journal* (2016), doi: <http://dx.doi.org/10.1016/j.ccej.2016.03.112>

This is a PDF file of an unedited manuscript that has been accepted for publication. As a service to our customers we are providing this early version of the manuscript. The manuscript will undergo copyediting, typesetting, and review of the resulting proof before it is published in its final form. Please note that during the production process errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

Removal of antibiotic compounds by adsorption using glycerol-based carbon materials

S. Álvarez-Torrellas^{†1}, R.S. Ribeiro², H.T. Gomes², G. Ovejero¹, J. García^{1†}

(1) - Grupo de Catálisis y Procesos de Separación (CyPS), Departamento de Ingeniería Química, Facultad de Ciencias Químicas, Universidad Complutense de Madrid, Avda. Complutense s/n, 28040 Madrid, Spain.

(2) - Laboratory of Separation and Reaction Engineering - Laboratory of Catalysis and Materials (LSRE-LCM), Department of Chemical and Biological Technology, School of Technology and Management, Polytechnic Institute of Bragança, Campus de Santa Apolónia, 5300-857 Bragança, Portugal.

Abstract

This study is focused on the synthesis and application of glycerol-based carbon materials (GBCM₂₀₀, GBCM₃₀₀ and GBCM₃₅₀) as adsorbents for the removal of the antibiotic compounds flumequine and tetracycline from aqueous solution. The synthesis enrolled the partial carbonization of a glycerol-sulphuric acid mixture, followed by thermal treatments under inert conditions and further thermal activation under oxidative atmosphere. The textural properties were investigated through N₂ adsorption-desorption isotherms, and the presence of oxygenated groups was discussed based on zeta potential and Fourier transform infrared (FTIR) data. The kinetic data revealed that the equilibrium time for flumequine adsorption was achieved within 96 h, while for tetracycline, it was reached after 120 h. Several kinetic models, i.e., pseudo-first order, pseudo-second order, fractional power, Elovich and Weber-Morris models, were applied, finding that the pseudo-second order model was the most suitable for the fitting of the experimental kinetic data. The estimated surface diffusion coefficient values, D_s , of 3.88 and $5.06 \times 10^{-14} \text{ m}^2 \cdot \text{s}^{-1}$, suggests that the pore diffusion is the rate limiting step of the adsorption process. Finally, as it is based on SSE values, Sips model well-fitted the experimental FLQ and TCN adsorption isotherm data, followed by Freundlich equation. The maximum adsorption capacities for flumequine and tetracycline was of 41.5 and $58.2 \text{ mg} \cdot \text{g}^{-1}$ by GBCM₃₅₀ activated carbon.

Keywords: Adsorption; antibiotics; carbon materials; emerging contaminants

[†]Corresponding author: Tel.: +34-91-394-5207; Fax: +34-91-394-4114.

E-mail address: satorrellas@ucm.es; jgarciar@ucm.es

1. Introduction

In the past few decades, the presence of several organic compounds has been detected at trace concentrations (ng.L^{-1} to $\mu\text{g.L}^{-1}$) in many water sources. New advances in analytical instrumentation have allowed the detection of these substances, now known as emerging contaminants, included in different categories, such as surfactants, pharmaceutically active compounds (PACs), personal care products, endocrine disruptor compounds (EDCs), pesticides, illicit drugs, gasoline additives, flame retardants, polychlorinated biphenyls (PCBs), polycyclic aromatic hydrocarbons (PAHs), perfluorinated compounds, dioxins and hormones, among others.

An important characteristic of these compounds is that they are continuously introduced into the environment, compensating the transformation/biodegradability rates [1]. In general, these pollutants are persistent, bioaccumulative and potentially toxic at medium to long term [2].

Among this group of substances, antibiotics have raised a significant interest from the scientific community. Sköld [3] defines the antibiotics as medicines that act selectively on bacteria, among them the pathogens, while leaving human cells and tissues unaffected. Antibiotics have been used for decades in the treatment of infections and related diseases. More than 250 different antibiotics are registered for human and veterinary uses [4]. It has been reported that most of the antibiotics are not adequately digested by organisms, leading to the excretion of these substances and their metabolites through the excrements and urine. In addition, the antibiotic residues –metabolites- in the environment are of great concern due to can induce antibiotic resistance genes at relatively low concentrations [5]. In addition, several scientific studies have demonstrated that the exposure to antibiotics (in $\mu\text{g.L}^{-1}$ to mg.L^{-1} order of magnitude) may also cause significant impact on aquatic organisms [6-8]. So, this is an issue of increasing concern, since the amount and variety of antibiotics in urban wastewater effluents, surface and groundwaters is growing every year [9].

Flumequine ($\text{C}_{14}\text{H}_{12}\text{FNO}_3$), labelled further in the text as FLQ, is an antibiotic belonging to the fluoroquinolones group, potent broad spectrum antibacterial agents, with an activity focused specifically to *Gram negative* bacteria. Flumequine, classified as a fluorinated compound (Group III, tricyclic derivatives), is characterized by a 9-fluorine atom, resulting in a 10-fold increase in activity against bacteria compared to nalidixic acid. Flumequine is widely used in veterinary medicine to control bacterial infections, being present in farms effluents [10-11], but it has also been detected in

hospital effluents [9]. So, persistent antibacterial agents will enhance unfavorable environmental effects. Sediment and fish pathogenic bacteria isolated from sediment fish farms have been found to develop resistance after exposure to antibiotic wastes, even 4-fold higher than the minimum inhibitory concentration [12]. Due to its low biodegradability, fluoroquinolone drugs are frequently detected in raw water, sediments and sand filtration [13]. The presence of flumequine has been detected in concentrations from 51 to 261 $\mu\text{g.kg}^{-1}$ in sediments and soils [14], and up to 578.8 $\mu\text{g.kg}^{-1}$ in fish farms and their aquatic medium [15-16]. Furthermore, flumequine has also already been detected in surface waters with concentrations up to 32 ng L^{-1} in France, which raises particular concerns about its possible reintroduction in the human consumption circuit [9]. At the same time, it has been detected significant LC_{50} values of flumequine on bacteria, crustaceans and aquatic fauna [17].

On its turn, tetracycline ($\text{C}_{22}\text{H}_{24}\text{N}_2\text{O}_8$), referenced as TCN, belongs to the tetracyclines group, a broad spectrum antimicrobial activity family of antibiotics, which is one of the most commonly used in human and veterinary therapies [18]. Tetracycline possesses a planar structure with four fused rings, with hydrophilic groups in one face and hydrophobic in the other face, including groups such as phenol, alcohol, ketone and amino. This compound has been specifically identified as one of the most developers of resistance in ecosystems. Tetracycline concentrations up to 10.0 $\mu\text{g.L}^{-1}$ have been detected in sediments near to poultry slaughterhouses and higher concentrations of tetracycline metabolites have been measured, up to 25.0 $\mu\text{g.L}^{-1}$ [19]. Furthermore, tetracycline has also been found in several urban wastewater effluents (in the range of 3.5-1420 ng L^{-1}); likewise, a concentration up to 80 ng L^{-1} has been detected in Australian surface waters [9]. Consequently, the removal of the antibiotics flumequine and tetracycline from aquatic environments is of great importance at the societal level. Therefore, it poses a challenge for the current research on new wastewater treatment options.

Adsorption, since it is a non-destructive tertiary process, has been revealed as one of the most advantageous physicochemical technique in wastewater treatment processes. In the research practice, it is very important to find a material showing high binding affinity towards the contaminant, and high selectivity and adsorption capacity properties. Commercial activated carbons possess most of these features, since they show a desirable affinity to hundreds of organic and inorganic compounds [20-21]. Since the pore diffusion seems to be the controlling mechanism in activated carbons

adsorption, the use of commercial adsorbents is usually associated with high expensive and environmental undesirable regeneration operations.

Crude glycerol, obtained from the biodiesel production, is actually an abundant feedstock. So, the global biodiesel market is estimated to reach 37 billion gallons by 2016 with an average annual growth of 42%, which means about 4 billion gallons of crude glycerol will be produced [22-23]. Thus, the preparation of carbonaceous materials by partial carbonization and sulfonation of this by-product consists of an interesting research focus. Bearing this in mind, the aim of this study is to explore several types of glycerol-based carbon materials synthesized by partial carbonization of glycerol in concentrated sulphuric acid solution for the removal of flumequine and tetracycline from aqueous solutions. In addition, an extensive study on the determination of the mass transfer coefficients was developed.

As far as we know, there are not previous studies on the removal of antibiotics from water by carbonaceous materials prepared from glycerol [24]. So, this is an innovative work on the adsorption of flumequine and tetracycline onto carbons prepared by carbonization of glycerol.

2. Materials and methods

2.1. Materials.

Sulphuric acid (95-97 wt.%) was obtained from Panreac and glycerol (99 wt.%) was purchased from Alfa Aesar. Antibiotics, flumequine (FLQ) and tetracycline (TCN) were purchased from Sigma-Aldrich, showing a purity higher than 98%. All the chemicals were used as received, without further purification. Ultrapure water was used in all the experiments.

The physico-chemical properties of flumequine and tetracycline are reported in Table 1.

[Table 1]

2.2. Synthesis procedure of glycerol-carbon materials.

Glycerol-based carbon materials were prepared adapting the procedure described in the work of Ribeiro et al. [30]. A mixture of glycerol (10 g) and concentrated (95-97 wt.%) sulphuric acid (40 g) was gently heated to 180 °C and left at that temperature for 20 min to allow in situ partial carbonization. The resulting material was ground in a

ceramic mortar to obtain particle sizes in the range 0.425-0.600 mm and further calcined in a tube furnace under a nitrogen flow ($100 \text{ cm}^3 \cdot \text{min}^{-1}$) at 120°C , 400°C and 600°C during 60 min at each temperature, and finally at 800°C for 240 min, defining a heating rate of $2^\circ\text{C} \cdot \text{min}^{-1}$. The as-calcined material was named GBCM and showed a basic character, due to the decomposition of the sulphonic groups during the calcination process. Then, the GBCM material was thermally activated in a tube furnace under oxidative atmosphere (air flow of $100 \text{ cm}^3 \cdot \text{min}^{-1}$) during 60 min at different temperatures, 200°C , 300°C and 350°C , resulting in materials with different textural and chemical properties. The carbon materials were labelled as GBCM followed by a subscript number corresponding to the activation temperature; e.g., GBCM₂₀₀, GBCM₃₀₀ and GBCM₃₅₀. Scheme 1 show the synthesis of GBCM. It is noteworthy that studies in continuous mode are envisaged in the following of this work, this being the reason why larger particle sizes were selected when compared to the ranges typically employed in slurry experiments.

[Scheme 1]

2.3. Characterization of the carbon materials

Textural characterization of the activated carbons was obtained from N_2 adsorption isotherms at -196°C , using a Micromeritics ASAP 2020 apparatus. Specific surface area (S_{BET}) of the materials was determined using BET equation. Total micropore volume (V_{Mic}) was calculated by Dubinin-Radushkevich equation and the total pore volume (V_{Total}) was determined at $P/P^0 = 0.95$. The pore size distribution was estimated using the Density Functional Theory (DFT) method. In DFT analysis it was assumed a slit pore geometry and a pressure range from $P/P^0 = 0.00$ to $P/P^0 = 0.90$ when the calculation assignments are not used.

Chemical surface characterization was performed by using Fourier transformed infrared spectrum (FTIR), recorded in the range $400\text{-}4600 \text{ cm}^{-1}$ and using a Thermo Nicolet FTIR spectrophotometer. Raman analysis of the samples was conducted by a Ntegra Spectra (NT-MDT) micro-Raman equipment, equipped with a CCD Andor iDus detector. The laser power and the wavelength at the sample surface was controlled at 20 mW and 532 nm, respectively. The samples were focalized by using an Olympus BXFM microscope.

The isoelectric point (pH_{IEP}) was measured in a Zeta sizer Nano ZS equipment, using 0.05 g of sample suspended in 20 mL of deionized water, the pH being adjusted with either 0.1 M HCl or 0.1 M NaOH solutions. The zero potential point of the sample was calculated by interpolation method. Finally, the composition of the carbon materials was determined by elemental analysis, in a LECO CHNS-932 elemental micro-analyzer.

2.4. Kinetic and equilibrium adsorption experiments

Both adsorption kinetic and equilibrium studies were carried out in an orbital shaker, by contacting different weights of adsorbent with 1.5 mL of FLQ or TCN solutions ($C_0 = 100 \text{ mg.L}^{-1}$). Temperature and shaking were controlled at $30 \pm 1^\circ\text{C}$ and 300 rpm, respectively.

Samples were filtered before analysis in 0.45 μm nylon filters. FLQ concentration was measured in a Varian Prostar high performance liquid chromatograph equipped with a UV-vis *diode array* detector. The measurements were performed in a Mediterranea C18 column 250 mm x 4.6 mm x 5 μm (Teknokroma), using a mobile phase of acetonitrile:water (0.1 %v H_3PO_4) (60:40 v/v) at a flow rate of 1.0 mL.min^{-1} and using a *loop* volume of 20 μL . TCN concentration measurements were accomplished in a Perkin Elmer Lambda 35 UV-vis spectrophotometer. The maximum wavelength used in the detection was 356 nm. All the FLQ and TCN samples were analysed in duplicate.

3. Results and discussion

3.1. Characterization of the carbon materials

Textural properties, e.g., specific surface area (S_{BET} , $\text{m}^2.\text{g}^{-1}$), total pore volume (V_{Total} , $\text{cm}^3.\text{g}^{-1}$), and micropore volume (V_{Mic} , $\text{cm}^3.\text{g}^{-1}$) of the carbonaceous materials are shown in Table 2. The N_2 adsorption-desorption isotherms of the tested carbonaceous materials are shown in Figure 1. GBCM is a non-porous material ($S_{\text{BET}} = 10 \text{ m}^2.\text{g}^{-1}$ and $V_{\text{Total}} = 0.02 \text{ cm}^3.\text{g}^{-1}$). The thermal activation allowed the development of porosity in the carbon materials, GBCM₂₀₀, GBCM₃₀₀ and GBCM₃₅₀. From textural properties (Table 2), it can also be observed the effect of the different thermal treatments on the specific surface area of glycerol-based carbon materials. These results are in agreement to those previously reported for similar materials, but with smaller particles sizes [30], since higher activation temperatures led to an increase in the specific surface area. Although all the glycerol-activated carbons possess a strong microporous nature, with low non-

microporous surface areas, the increase in the thermal activation temperature affects the micropore volume, with the largest micropore volume being obtained in the sample activated at the highest temperature ($0.217 \text{ cm}^3.\text{g}^{-1}$ for GBCM₃₅₀ carbon).

[Figure 1]

[Table 2]

Pore size distribution of the tested materials was evaluated by Density Functional Theory method. Pore size of the carbonaceous materials (Table 2) confirmed that the thermal treatment temperature had an effect on the pore size of the glycerol-based carbon materials, showing values from 0.86 to 1.21 nm, as the treatment temperature increased from 200 °C to 350 °C.

SEM micrographs, developed in order to study the morphology of the carbons, are shown in Figure 2. The micrographs show smooth carbon surfaces, revealing fractures in the solid surface, which may arise from the grinding process; both GBCM₃₀₀ and GBCM₃₅₀ show quite similar surface morphology, and the presence of pores dispersed on their surface.

[Figure 2]

FT-IR spectra of the tested activated carbons in the wavelength range from 400 to 4000 cm^{-1} are shown in Figure 3. FTIR spectrum defines the active chemical groups present on the carbon surfaces. A broad band can be found at 3400 cm^{-1} , attributed to the O-H stretching vibrations, characteristic of the presence of surface hydroxylic groups and chemisorbed water.

The peak at 2920 cm^{-1} in all the FT-IR spectra is characteristic of C-H stretching vibrations, suggesting the presence of alkane/alkene groups on the activated carbons surface [31]. The absorption band at 1645 cm^{-1} , showing high intensity in glycerol-based carbon materials spectra, is assigned to C=O stretching vibrations of carboxylic, ester, lactones, quinone, diketone, ketoester and keto-enol structures [31-32].

The sharp peaks at 1384 cm^{-1} in the glycerol-based carbon materials spectra indicates S=O stretching vibrations, due to the presence of sulfate or sulfonyl groups, which are expected since sulphuric acid was used in the synthesis of these materials.

The absorption band at 1118 cm^{-1} is usually ascribed to -OH and/or C-O stretching vibrations of alcohol, carboxylate, ester and/or ether groups [32].

[Figure 3]

It is well-known that Raman spectroscopy is the nondestructive and most direct technique to characterize the structure of carbon-based materials. The G band at a wavelength about 1580 cm^{-1} is a peak attributed to the stretching of all sp^2 atomic pairs on carbon ring or long-carbon chain in graphite. This band can be used to determine the degree of symmetry and order of the graphitic structure. The spectrum of the ordinary carbon materials exhibits an additional first-order band (D band) at around 1360 cm^{-1} due to the disordered arrangement and low symmetry of graphite lattice structure. It has been established that the D band resulted from the structural defection created by the attachment of hydroxyl and epoxy groups on the carbon basal plane. The value of R , expressed in Eq. 1, is used to characterize the graphitization degree, reflecting the structural proximity degree of the carbon material and the ideal graphite structure. I_D and I_G were calculated from Gaussian peak fitting function, representing the peak areas of band D and band G, respectively.

$$R = I_D / I_G \quad (1)$$

The Raman spectra of the tested glycerol-based carbon materials are shown in Figure 4. The two peaks at around $1600\text{-}1610\text{ cm}^{-1}$ and $1365\text{-}1370\text{ cm}^{-1}$, corresponding to the G and D bands, could be found for all the three samples, indicating similar graphite-like crystallite and amorphous carbon structure of the samples. R values (Table 3) increased as the activation temperature did it, suggesting that the graphite crystallize size of the carbon samples decreased due to the increasing in the unsaturated carbon atoms in the corners and edges of the carbon structure, this is, when the oxygenated functionalities were incorporated to the carbon surface. Similar findings were reported in the literature [33]. A 2D peak (2800 cm^{-1}) can be observed, more clearly detected for GBCM_{350} carbon material. It could be associated to the disposition of the graphitic structure, related to the substituted groups, i.e., alkane, alkene and/or alkyne groups [34].

[Figure 4]

The isoelectric point (pH_{IEP}) of the carbonaceous adsorbents was determined using zeta potential measurements. The results are shown in Table 3. The glycerol-based carbon material activated at 200 °C, GBCM₂₀₀, possesses a pH_{IEP} value of 5.6, and as the activation temperature increased, the oxygenated functionalities of the materials also increased, as suggested by the isoelectric points of GBCM₃₀₀ and GBCM₃₅₀ carbons (pH_{IEP} = 3.8 and 2.6, respectively).

Elemental analysis was performed in order to determine the chemical composition (%C, %H, %N, %S and %O) of the carbonaceous samples (Table 3). These results are in accordance to the acidic/basic character of the samples as inferred by the isoelectric point measurements. As expected, the results given in Table 3 suggest that the acidic character of the carbonaceous materials is mainly due to the incorporation of oxygen-containing surface groups during the thermal activation in air [35]; so, carbon treated at the highest temperature, GBCM₃₀₀, is more hydrophilic than GBCM₂₀₀ and GBCM₃₅₀.

[Table 3]

3.2. Adsorption kinetic studies

The evolution of FLQ and TCN adsorption versus time onto GBCM₃₀₀ carbon material is illustrated in Figure 5. FLQ equilibrium time was achieved within 96 hours, while for TCN it was reached after 120 hours. Generally, an initial faster increasing in the adsorption capacity was observed during first 6-10 hours, approximately. In this behaviour, it is involved the external surface area of the carbons, supporting the access of the molecules to the inner porous structure. So, the faster kinetic range is directly related to the contribution of the external area of the material.

Afterwards, the adsorption capacity slowly increased until the equilibrium was reached. The molecular dimensions and the average pore size of the carbon are intimately related, in order to contribute to the kinetic and adsorption capacity values. FLQ molecule dimensions (1.0x1.0x0.38 nm) are very close to the average pore width of GBCM₃₀₀ material (Table 2), so it can be inferred that the molecule could be adsorbed in the inner narrow pores. It is established that compounds like FLQ are more likely to adsorb in pores approximately the same size as the adsorbate, where there will be more contact points and therefore more favourable adsorption energy [36]. Otherwise, TCN molecule, due to its planar and linear configuration (1.29x0.62x0.75

nm), have to be oriented parallel to the inner pore walls [37], as a cylinder, in order to facilitate the access to the more narrow cavities.

In addition to the contribution of external and specific surface area of the adsorbent to the molecules adsorption, some specific interactions may have been established, since pseudo-second order kinetic model provided better fitting results than pseudo-first order model. This is according to previous research works [38-39], which state that the successful application of the pseudo-second order kinetic model could suggest that chemisorption, e.g., binding interactions, is occurring.

[Figure 5]

3.2.1. Kinetic modelling

With the purpose of studying the kinetic adsorption behavior of FLQ and TCN onto GBCM₃₀₀, several kinetic models have been used for the fitting of the experimental data.

The fractional power function model is an empirical model that describes the relation between the mass of the adsorbate per unit mass of the adsorbent and time 't' [40]. It is as follows:

$$q = a \cdot t^b \quad (2)$$

where, q (mg.g⁻¹) is the adsorption capacity of flumequine/tetracycline at given time, a (mg.g⁻¹.h⁻¹) and b are adjustment parameters. The linearized form of the power function equation is given as:

$$\ln q = \ln a + b \cdot \ln t \quad (3)$$

When b = 0.5, the equation becomes the same as the Weber-Morris intraparticle diffusion model, discussed further.

Lagergren pseudo-first order kinetic model is one of the most frequent equations used to describe the adsorption of different adsorbates from aqueous solution onto an adsorbent [41-42]. It could be written as the following equation:

$$q = q_e \cdot (1 - e^{-k_1 \cdot t}) \quad (4)$$

In order to determine the model rate constant, it is usually expressed in its linearized form:

$$\ln(q_e - q) = \ln q_e - k_1 \cdot t \quad (5)$$

where, q_e , mg.g^{-1} , and q , mg.g^{-1} , are equilibrium adsorption capacity and adsorption capacity at time t , respectively; k_1 , h^{-1} , is the pseudo-first-order model rate constant.

The pseudo-second order equation [43] has also been considered to explore the kinetic adsorption of the antibiotics onto the glycerol-based carbon material. It is given as the next equation:

$$q = \frac{q_e^2 \cdot k_2 \cdot t}{1 + q_e \cdot k_2 \cdot t} \quad (6)$$

The linearized form of the pseudo-second order model is given as Equation 7:

$$\frac{t}{q} = \frac{1}{k_2 \cdot q_e^2} + \frac{t}{q_e} \quad (7)$$

where, q_e , mg.g^{-1} , and q , mg.g^{-1} , are equilibrium adsorption capacity and adsorption capacity at time t , respectively; and k_2 , $\text{g.mg}^{-1}.\text{h}^{-1}$, is the pseudo-second-order adsorption rate constant.

Finally, the kinetic model proposed by Elovich was used. The Elovich equation assumes that the active sites of the adsorbent are heterogeneous and therefore exhibit different activation energies for chemisorption [44-45]. Teng and Hsieh [46] proposed that constant α is related to the rate of chemisorption and β is associated to the surface coverage. The Elovich model is presented by the following equation:

$$\frac{dq}{dt} = \beta \cdot e^{-\alpha \cdot q} \quad (8)$$

$$q = \beta \cdot \ln(\alpha \cdot \beta \cdot t) \quad (9)$$

where q , mg.g^{-1} , is the adsorption capacity at time t , α ($\text{g.mg}^{-1}.\text{h}^{-1}$) is the initial adsorption rate and β ($\text{mg.g}^{-1}.\text{h}^{-1}$) is the desorption constant.

The linearized form of Elovich model can be written as:

$$q = \beta \cdot \ln(\alpha \cdot \beta) + \beta \cdot \ln t \quad (10)$$

The parameters obtained for the kinetic models applied and the regression coefficients, R^2 , are shown in Table 4.

[Table 4]

The goodness of the fitting to the experimental data was additionally studied through the non-linearized forms of the kinetic equations, minimizing the sum of the

squares of the residuals (SSE) (Eq. 11) [47]. The parameters obtained from non-linearized equations can be seen in Table 5 and the experimental and theoretical data are depicted in Figure 6.

$$SSE = \sum (q_{\text{exp}} - q_{\text{cal}})^2 \quad (11)$$

Considering R^2 values, it can be observed that all the models could reasonably fit the experimental data. Lower correlation values obtained for Lagergren pseudo-first order model showed that there is no prominent relationship between that model and experimental kinetic data. In the opposite, it was observed an accurately fitting of pseudo-second order equation to the experimental FLQ and TCN adsorption data. The non-linearized fitting to the experimental data revealed good results for pseudo-second order model, since better fitting was obtained by applying fractional power model to TCN adsorption data and both pseudo-second order and Elovich models to FLQ experimental adsorption data. In general, better predictions were obtained by using the non-linearized equations of the kinetic models (Fig. 6).

Similar kinetic findings have been reported in the literature, such as the removal of atrazine and paracetamol onto activated carbons [48], or the depletion of estrogenic compounds from aqueous solutions using carbon nanotubes [49].

Half-adsorption time, $t_{1/2}$, parameter determined by linearized pseudo-second order equation and defined as the time needed to reach 50% of saturation of the adsorbent, was found to be 8.10 and 14.87 h for FLQ and TCN, respectively.

In addition, the FLQ and TCN initial adsorption rates (v_0) onto GBCM₃₀₀ were estimated in 2.00 and 1.27 mg.g⁻¹.h⁻¹. So it seems that FLQ adsorption process in the initial stage is faster than TCN removal [50].

[Table 5]

[Figure 6]

The high accuracy in the prediction of experimental data by pseudo-second order model has been considered in many research works [38-39] as an evidence of that chemical bindings are involved in the adsorption mechanism of these antibiotics onto GBCM₃₀₀. Therefore, this observation cannot be demonstrated only through the development of experimental adsorption tests and their fitting, since strong interactions

could be misinterpreted as molecules trapped within the pore network. The study of the FTIR spectra of the adsorbed phase onto GBCM₃₀₀ (Fig. 7) confirmed this assumption, not revealing strong interactions between both the adsorbent and the adsorbed phase. The FTIR analysis of FLQ and TCN adsorbed onto the carbon showed significance similarity with the original spectra, sifted peaks, or changes in the intensity of the transmittance peaks could not be observed.

[Figure 7]

3.2.2. Adsorption mechanistic studies. Intra-particle diffusion model.

Generally, three stages can be distinguished in an adsorption process:

- First step: the adsorbate mass transfer from the aqueous phase to the more external adsorbent surface; film diffusion phenomenon is occurring.
- Second step: internal diffusion of adsorbate within the more microporous structure of the adsorbent, named as particle diffusion.
- Third step: adsorption of the adsorbate onto the surface sites. Because this stage is very fast in nature, it cannot be considered as a rate limiting step.

In order to investigate the mechanisms governing the adsorption process and the rate limiting steps involved in the removal of the antibiotics under study, the intra-particle diffusion model (expressed by Weber and Morris plot) was applied to the experimental FLQ and TCN adsorption data. It was expressed by the following equation:

$$q = k_i \cdot t^{1/2} + C_i \quad (12)$$

where, k_i ($\text{mg} \cdot \text{g}^{-1} \cdot \text{h}^{-0.5}$) is the intra-particle diffusion rate constant of the stage i , which is calculated through the slope of the straight-line portion of plotting q vs. $t^{1/2}$, as it is shown in Figure 8. C_i parameter gives an idea about the thickness of the boundary layer; this is, as the larger value of C_i , greater is the boundary layer effect in the adsorption process [51].

Though the experimental data depicted in Fig. 8 passed through the origin, indicating that there is no boundary layer effect, the plot showed three differenced phases, suggesting that pore diffusion is not only the rate-controlling step and other mechanisms are also involved in the adsorption process.

[Figure 8]

For both contaminants, a better correlation in the determination of the rate constant, $k_{i,2}$, of Weber and Morris model (Table 6) was found, indicating that the intra-particle diffusion mechanism is appropriated to describe the adsorption process of FLQ and TCN onto GBCM₃₀₀. This finding can be related to the highly microporous nature of GBCM₃₀₀ carbon material (Table 2).

[Table 6]

3.2.3. Estimation of the film mass transfer coefficient.

From the general bulk-phase mass balance equation (Eq. 13) it can be demonstrated that the adsorption rate is related with the film mass transfer coefficient, k_f , [52] as described by Eq. 14.

$$V \cdot (C_0 - C) = m \cdot q \quad (13)$$

where, C_0 and C ($\text{mg} \cdot \text{L}^{-1}$) are respectively the initial adsorbate concentration and the adsorbate concentration at time t in the solution, V (L) is the volume of solution, m (g) is the mass of adsorbent and q ($\text{mg} \cdot \text{g}^{-1}$) is the adsorption capacity of the system.

$$\frac{dq}{dt} = -\frac{V}{m} \cdot \frac{dC}{dt} = \frac{3 \cdot k_f}{R \cdot \rho_p} \cdot (C - C_s) \quad (14)$$

where, C_s ($\text{mg} \cdot \text{L}^{-1}$) is the liquid phase concentration at the adsorbent surface.

Initially ($t = 0$), $C = C_0$ and $C_s = 0$, and the initial adsorption rate, r_0 , is defined as:

$$r_0 \equiv \left(\frac{dq}{dt} \right)_{t=0} = \frac{3 \cdot k_f \cdot C_0}{R \cdot \rho_p} \quad (15)$$

Substituting in Equation 15,

$$k_f = \frac{R \cdot \rho_p \cdot r_0}{3 \cdot C_0} \quad (16)$$

where, R (m) is the adsorbent particle radius, and ρ_p ($\text{g} \cdot \text{L}^{-1}$) is the adsorbent particle density.

This expression allows to calculating the film mass transfer coefficient through the initial adsorption rate. Here in this work, it has been proposed the estimation of the initial adsorption rate based on the research of Yao and Chen [53]. The calculation considers the early adsorption stage, corresponding to the range of adsorption capacity

data from zero to the approximately half of the equilibrium adsorption capacity. Thus, the early-stage kinetic equation can be expressed as:

$$q = \frac{r_0 \cdot t}{1 + a \cdot t} \quad (17)$$

where, r_0 ($\text{mg} \cdot \text{g}^{-1} \cdot \text{h}^{-1}$) is the initial adsorption rate and a is a parameter that can be determined through a linear regression (Eq. 18), by plotting t/q versus t .

$$\frac{t}{q} = \frac{1}{r_0} + \frac{a}{r_0} \cdot t \quad (18)$$

The expression in Eq. 18 is similar to the pseudo-second order kinetic equation; the difference is that the parameters used here are only for the early adsorption stage of the experimental kinetic data.

3.2.4. Estimation of surface diffusion coefficient.

Equation 13 can be rewritten as:

$$V \cdot (C - C_e) = m \cdot (q_e - q) \quad (19)$$

and the surface diffusion coefficient, D_s , can be defined as follows:

$$\frac{dq}{dt} = \frac{\left[1 + \frac{m}{V} \cdot \left(\frac{dq}{dC}\right)_e\right] \cdot (q_e - q)}{\frac{R^2}{\pi^2 \cdot D_s} + \frac{R \cdot \rho_p}{3 \cdot k_f} \cdot \left(\frac{dq}{dC}\right)_e} \quad (20)$$

In order to define the near to equilibrium or late-stage adsorption rate constant, k_e , it is proposed the next expression:

$$k_e = \frac{1 + \frac{m}{V} \cdot \left(\frac{dq}{dC}\right)_e}{\frac{R^2}{\pi^2 \cdot D_s} + \frac{R \cdot \rho_p}{3 \cdot k_f} \cdot \left(\frac{dq}{dC}\right)_e} \quad (21)$$

If the Eq. 21 is applied only to the late or near-equilibrium adsorption stage, it can be rewritten as:

$$\frac{1}{k_e} = \frac{1}{1 + \frac{m}{V} \cdot \left(\frac{dq}{dC}\right)_e} \cdot \frac{R^2}{\pi^2 \cdot D_s} + \frac{1}{1 + \frac{m}{V} \cdot \left(\frac{dq}{dC}\right)_e} \cdot \frac{R \cdot \rho_p \cdot \left(\frac{dq}{dC}\right)_e}{3 \cdot k_f} \quad (22)$$

where, the left side of the equation is representing the mass transfer resistance in the late adsorption stage and the two terms on the right side are the surface diffusion, D_s , and the film mass transfer resistance, k_f , respectively.

Equation 21 can be rearranged in order to determine the surface diffusion coefficient:

$$D_s = \frac{(R/\pi)^2}{\frac{1}{k_e} \cdot \left[1 + \frac{m}{V} \cdot \left(\frac{dq}{dC} \right)_e \right] - \frac{R \cdot \rho_p}{3 \cdot k_f} \cdot \left(\frac{dq}{dC} \right)_e} \quad (23)$$

The method for determining the k_e parameter, similar to that of the calculation of r_0 , is proposed by the next linear equation:

$$-\ln \left(1 - \frac{q}{q_e} \right) = b + k_e \cdot t \quad (24)$$

where, k_e and b can be estimated by plotting the experimental kinetic data corresponding to the range of q from about 70% to nearly 100% of the equilibrium adsorption capacity (q_e).

The calculated values of r_0 , k_e , k_f and D_s are given in Table 6. For FLQ and TCN adsorption, surface diffusion can be considered the rate-controlled stage, since D_s took very much lower values than the film mass transfer coefficients, i.e., $D_s = 3.88 \times 10^{-14}$ and $5.06 \times 10^{-14} \text{ m}^2 \cdot \text{s}^{-1}$, for FLQ and TCN, respectively. Film diffusion and particle diffusion coefficients in the order of magnitude of 5.1×10^{-8} and $8.6 \times 10^{-9} \text{ cm}^2 \cdot \text{s}^{-1}$ were reported in the literature for the adsorption of tetracycline onto nanocellulose [54].

3.4. The effect of pore structure on the adsorption capacity.

Generally, in the removal of organic compounds by carbonaceous materials, it is well known that the specific surface area and pore size distribution of the adsorbents highly control the adsorption behavior. FLQ and TCN experimental adsorption capacities were plotted versus the specific surface area, S_{BET} , of the carbon materials (Fig. 9), in order to investigate if there is any correlation between both parameters. Even though the BET equation may not offer accurate results in the calculation of the specific surface area of highly microporous materials, the values of FLQ adsorption capacity showed an acceptable linear correlation to the S_{BET} parameter, while the amount of TCN adsorbed do not seem to be influenced by the surface area of the materials. Bear this in mind, it may be concluded that the textural properties could affect the extent of FLQ adsorption removal. In the opposite, other interactions, such as π - π EDA (electron-donor-acceptor) with the graphite structure, seem to highly condition the adsorption

removal of TCN by the carbon materials, as previously reported in the study of TCN removal onto several carbonaceous and clays materials [55-56].

[Figure 9]

3.5. Adsorption isotherms. Equilibrium modelling.

Both FLQ and TCN experimental adsorption isotherms (Figures 10 a-b) showed two-step or three-step adsorption profiles, and could be classified as S-3 type, according to Giles classification [57]. Accordingly, important information about the adsorbate-adsorbent interactions could be withdrawn from the multistep-shaped isotherms. Maybe this sigmoid trend could be attributed to a synergistic effect, which is favouring the adsorption of FLQ or TCN molecules by adsorbate-adsorbate interactions to the adsorbed molecules. The sudden increasing in the adsorption capacity at higher aqueous concentration could be related to a more favourable reorientation of the adsorbate molecules, generating more availability for adsorption in the active sites. Similar trend in the adsorption isotherms was found by Gómez et al. [58] in the removal of acid dyes by activated carbon and in the work of Konda et al. [59], studying the sorption of diazinon, isoproturon, atrazine and imidacloprid on soil.

[Figure 10]

From Fig. 10 a-b can be stated that FLQ removal was highly influenced by the textural properties of the carbonaceous materials, since GBCM₃₅₀ carbon, which offered the highest specific surface area among the tested adsorbents led to the maximum FLQ adsorption capacity, i.e., 41.5 mg.g⁻¹. In the opposite, TCN adsorption isotherms did not reveal a trending related to the textural properties of the materials; this is, GBCM₂₀₀, GBCM₃₀₀ and GBCM₃₅₀ carbons showed similar adsorption capacity values: 53.9, 51.2 and 58.2 mg.g⁻¹, respectively.

Figure 11 showed a comparison of the experimental FLQ and TCN adsorption capacities obtained by the tested carbonaceous materials.

[Figure 11]

In this work, the fitting of the experimental data to Langmuir, Freundlich and Sips models, widely applied to describe experimental adsorption isotherms, was carried out. According to the assumptions of Langmuir model, the adsorbate molecules are adsorbed onto the solid surface in a monolayer coverage without interaction between adsorbates, and supposing that the adsorbent surface is energetically uniform. Langmuir equation can be described as follows:

$$q_e = \frac{q_{sat} \cdot b \cdot C_e}{1 + b \cdot C_e} \quad (25)$$

where, q_e ($\text{mg} \cdot \text{g}^{-1}$) is the adsorption capacity at equilibrium time, C_e ($\text{mg} \cdot \text{L}^{-1}$) is the concentration of adsorbate in the aqueous phase at equilibrium, q_{sat} ($\text{mg} \cdot \text{g}^{-1}$) is the maximum adsorption capacity for complete monolayer coverage and b ($\text{L} \cdot \text{mg}^{-1}$) is the Langmuir constant related to the affinity to the binding sites.

On its turn, Freundlich model assumes heterogeneous surface energies, in which the energy varies as a function of the surface coverage. Freundlich model is expressed by the next equation:

$$q_e = K_F \cdot C_e^{1/n} \quad (26)$$

where, K_F ($\text{L} \cdot \text{g}^{-1}$) is a Freundlich constant regarded to the adsorption capacity, and n is a parameter related to the adsorption intensity.

The Sips model [60] is a combination of Langmuir and Freundlich equations, that is given as:

$$q_e = \frac{q_m \cdot (K_S \cdot C_e)^{1/n}}{1 + (K_S \cdot C_e)^{1/n}} \quad (27)$$

where, q_m ($\text{mg} \cdot \text{g}^{-1}$) is the maximum adsorption capacity, K_S ($\text{L} \cdot \text{mg}^{-1}$) is the affinity constant, and n is a Sips parameter related to the heterogeneity of the adsorption system.

Models parameters and SSE values (Eq. 11), determined by the using of the non-linearized theoretical models, are shown in Table 7. Accordingly, the experimental and theoretical adsorption data were compared in Figures 10 a-b.

Generally, Sips model with the lowest values of sum of square of residuals (SSE) showed the best fitting to both FLQ and TCN adsorption data, as compared to the other models. It could be found that Langmuir model was not a suitable equation for the prediction of flumequine and tetracycline removal onto the glycerol-based activated carbons. The good fitting of Sips isotherm model to the experimental data suggested

that adsorption is taking place both on homogeneous and heterogeneous surface of activated carbons, as it is referred in Tangsir et al. [61].

[Table 7]

4. Conclusions

In this work, several glycerol-based carbon materials for the removal of flumequine and tetracycline from aqueous solution have been developed. The obtained activated carbons exhibited high surface area, microporous structure and acidic character. It was found that the increasing in the thermal activation temperature increased both the porosity and the acidity of the materials, due to the incorporation of new oxygenated functionalities on the carbon surface.

Though the experimental kinetic data were well-fitted to the pseudo-second order kinetic model, the study of the FTIR spectra of the adsorbed phase onto GBCM₃₀₀ material revealed that strong chemical interactions with the adsorbent are not occurring. From the estimation of the film mass transfer coefficients, 1.09 and $1.97 \times 10^{-6} \text{ m.s}^{-1}$, and the surface diffusion coefficients, 3.88 and $5.06 \times 10^{-14} \text{ m}^2.\text{s}^{-1}$, for FLQ and TCN, respectively, it was suggested that the pore diffusion is the rate-controlling step of the process, in agreement to the results obtained by intra-particle diffusion model.

The adsorption of FLQ was found to be dependent on the textural properties of the carbon materials, with a maximum adsorption capacity of 41.5 mg.g^{-1} onto GBCM₃₅₀ activated carbon. FLQ adsorption capacities were found to be of 0.9 , 33.7 and 41.5 mg.g^{-1} , for GBCM₂₀₀, GBCM₃₀₀ and GBCM₃₅₀, respectively, while TCN values were of 53.9 , 51.3 and 58.1 mg.g^{-1} , for the same materials. The adsorption of both antibiotics could be well described by Sips isotherm model, confirming that the adsorption process was mainly multilayer. Then the synthesized materials could be a promising materials for environmental remediation, since they could be used for the removal of other antibiotic and/or organic compounds from water.

Acknowledgements

The authors would like to thank the financial support of Ministry of Economía and Competitividad of Spain (Contract TRAGUANET Network CTM2014-53485-REDC), the Regional Government of Madrid provided through Project REMTAVARES S2013/MAE-2716 and the European Social Fund. Additionally the research has been

partially supported by FCT and FEDER under Programme PT2020 (Project UID/EQU/50020/2013). R.S. Ribeiro acknowledges the FCT individual Ph.D. grant SFRH/BD/94177/2013, with financing from FCT and the European Social Fund (through POPH and QREN).

The authors also thank the anonymous referees for comments on this manuscript and to the Scientific Technological Centre (CAIS-ICTS, Complutense University) for providing the infrastructural support to carry out the Raman analyses.

References

- [1] D. Barceló, Emerging pollutants in water analysis, *Trends Anal. Chem.* 22 (2003) xiv-xvi.
- [2] C.G. Daughton, Non-regulated water contaminants: emerging research, *Environ. Impact Asses.* 24 (2004) 711-732.
- [3] O. Sköld, Antibiotics and Antibiotics Resistance, Ed. Wiley, New Jersey, 2011, pp. 207.
- [4] K. Kümmerer, Antibiotics in the aquatic environment, *Chemosphere* 75 (2009) 417-434.
- [5] G. Dantas, M.O. Sommer, R.D. Oluwasegun, G.M. Church, Bacteria subsisting on antibiotics, *Science* 320 (2008) 100-103.
- [6] K. Ji, S. Kim, S. Han, J. Seo, S. Lee, Y. Park, K. Choi, Y.L. Kho, P.G. Kim, J. Park, Risk assessment of chlortetracycline, oxytetracycline, sulfamethazine, sulfathiazole, and erythromycin in aquatic environment: are the current environmental concentrations safe?, *Ecotoxicology* 21 (2012) 2031-2050.
- [7] H.-T. Lai, J.-H. Hou, C.-I. Su, C.-L. Chen, Effects of chloramphenicol, florfenicol, and thiamphenicol on growth of algae *Chlorella pyrenoidosa*, *Isochrysis galbana*, and *Tetraselmis chui*, *Ecotoxicol. Environ. Saf.* 72 (2009) 329-334.
- [8] L. Wollenberger, B. Halling-Sørensen, K.O. Kusk, Acute and chronic toxicity of veterinary antibiotics to *Daphnia magna*, *Chemosphere* 40 (2000) 723-730.
- [9] D. Fatta-Kassinos, S. Meric, A. Nikolau, Pharmaceutical residues in environmental waters and wastewater: current state of knowledge and future research, *Anal. Bioanal. Chem.* 399 (2011) 251-275.
- [10] A. Bryskier, J. Chantot, Classification and structure-activity relationships of fluoroquinolones, *Drugs* 49 (1995) 16-28.

- [11] S. García-Segura, J.A. Garrido, R.M. Rodríguez, P.L. Cabot, F. Centellas, C. Arias, E. Brillas, Mineralization of flumequine in acidic medium by electro-Fenton and photoelectro-Fenton processes, *Water Res.* 46 (2012) 2067-2076.
- [12] H. Hektoen, J.A. Berge, V. Hormazábal, M. Yndestad, Persistence of antibacterial agents in marine sediments, *Aquaculture* 133 (1995) 175-184.
- [13] N.M. Vieno, H. Harkki, T. Tuhkanen, L. Kronberg, Occurrence of pharmaceuticals in river water and their elimination in a pilot-scale drinking water treatment plant, *Environ. Sci. Technol.* 41 (2007) 5077-5084.
- [14] M.D. Prat, D. Ramil, R. Compañó, J.A. Hernández-Arteseros, M. Granados, Determination of flumequine and oxolinic acid in sediments and soils by microwave-assisted extraction and liquid chromatography-fluorescence, *Anal. Chim. Acta* 567 (2006) 229-235.
- [15] G.M. Lalumera, D. Calamari, P. Galli, S. Castiglioni, G. Crosa, R. Fanelli, Preliminary investigation on the environmental occurrence and effects of antibiotics used in aquaculture in Italy, *Chemosphere* 54 (2004) 661-668.
- [16] J. Beausse, Selected drugs in solid matrices: a review of environmental determination, occurrence and properties of principal substances, *Trends Anal. Chem.* 23 (2004) 10-11.
- [17] B. Halling-Sørensen, S. Nors Nielsen, P.F. Lanzky, F. Ingerslev, H.C. Holten Lützhof, S.E. Jørgensen, Occurrence, fate and effects of pharmaceutical substances in the environment- A review, *Chemosphere* 36 (1998) 357-393.
- [18] S. Thrile-Bruhn, Pharmaceutical antibiotics compounds in soils –a review, *J. Plant Nutr. Soil Sci.* 166 (2003) 145-167.
- [19] E.I.A. Topal, Uptake of tetracycline and metabolites in *Phragmites australis* exposed to treated poultry slaughterhouse wastewaters, *Ecol. Eng.* 83 (2015) 233-238.
- [20] A. Zgajnar-Gotvaj, T. Tisler, J. Zagorc-Koncan, Comparison of different treatment strategies for industrial landfill leachate, *J. Hazard. Mater.* 162 (2009) 1446-1456.
- [21] J.L. Gao, V. Oloibiri, M. Chys, S. De Wandel, B. Decostere, W. Audenaert, Y.L. He, S.W.H. Van Hulle, Integration of autotrophic nitrogen removal, ozonation and activated carbon filtration for treatment of landfill leachate, *Chem. Eng. J.* 275 (2015) 281-287.
- [22] F. Yang, M.A. Hanna, R. Sun, Value-added used for crude glycerol-a byproduct of biodiesel production, *Biotechnol. Biofuels* 5 (2012) 1-10.

- [23] X. Fan, R. Burton, Y. Zhou, Glycerol (Byproduct of Biodiesel Production) as a source for fuels and chemicals- Mini Review, *Open Fuels Energy Sci. J.* 3 (2010) 17-22.
- [24] Sources of search: Web of Science; BIOSIS Citation IndexSM, BIOSIS Previews[®], Current Contents Connect[®], Derwent Innovations IndexSM, KCI-Korean Journal Database, MEDLINE[®], SciELO Citation Index. Keywords: glycerol-based carbon materials; adsorption; antibiotics. Date: 15th January of 2016.
- [25] J. Tolls, Sorption of veterinary pharmaceuticals in soils: A review, *Environ. Sci. Technol.* 35 (2001) 3397-3406.
- [26] C.T. Chiou, Reactions and movement of organic chemicals in soils, Eds. B.L. Sawhney, K. Brown. Soil Science Society of America: Madison, WI, 1989, pp. 1-30.
- [27] H. Ueda, R. Pereira-Rosario, C.M. Riley, J.H. Perrin, Some pharmaceutical applications of diode-array spectrophotometers, *Drug Dev. Ind. Pharm.* 11 (1985) 833-843.
- [28] G. Li, D. Zhang, M. Wang, J. Huang, L. Huang, Preparation of activated carbons from *Iris tectorum* employing ferric nitrate as dopant for removal of tetracycline from aqueous solutions, *Ecotox. Environ. Safe.* 98 (2013) 273-282.
- [29] F. Gambinossi, B. Mecheri, M. Nocentini, M. Puggelli, G. Caminati, Effect of the phospholipid head group in antibiotic-phospholipid association at water-air interface, *Biophys. Chem.* 110 (2004) 101-117.
- [30] R.S. Ribeiro, A.M.T. Silva, M.T. Pinho, J.L. Figueiredo, J.L. Faria, H.T. Gomes, Development of glycerol-based metal-free carbon materials for environmental catalytic applications, *Catal. Today* 240 (2015) 61-66.
- [31] P.E. Fanning, M.A. Vannice, A DRIFTS study of the formation of surface groups on carbon by oxidation, *Carbon* 31 (1993) 721-730.
- [32] B.K. Pradhan, N.K. Sandle, Effect of different oxidizing agent treatments on the surface properties of activated carbons, *Carbon* 37 (1999) 1323-1332.
- [33] Y. Zhang, G. Liu, S. Yu, J. Zhang, Y. Tang, P. Li, Y. Ren, Kinetics and interfacial thermodynamics of the pH-related sorption of tetrabromobisphenol A onto multiwalled carbon nanotubes, *ACS Appl. Mater. Interfaces* 6 (2014) 20968-20977.
- [34] J. Jiang, Y. Li, C. Gao, N.M. Kim, X. Fan, G. Wang, Z. Peng, R.H. Hauge, J.M. Tour, Growing carbon nanotubes from both sides of graphene, *ACS Appl. Mater. Interfaces* (2016) DOI: 10.1021/acsami.5b12254.
- [35] J.L. Figueiredo, Functionalization of porous carbons for catalytic applications, *J. Mater. Chem. A* 1 (2013) 9351-9364.

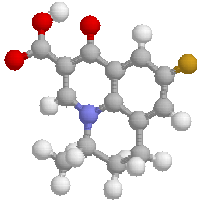
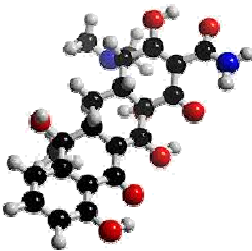
- [36] G. Newcombe, M. Drikas, R. Hayes, Influence of characterized natural organic matter on activated carbon adsorption: II. Effect on pore volume distribution and adsorption of 2-methylisoborneol, *Water Res.* 31 (1997) 1065-1073.
- [37] L.L. Ji, W. Chen, L. Duan, D.Q. Zhu, Mechanisms for strong adsorption of tetracycline to carbon nanotubes: a comparative study using activated carbon and graphite as adsorbents, *Environ. Sci. Technol.* 43 (2009) 2322-2327.
- [38] Y.S. Ho, G. McKay, The kinetics of sorption of divalent metal ions onto sphagnum moss peat, *Water Res.* 34 (2000) 735-742.
- [39] Y.S. Ho, G. McKay, Comparative sorption kinetic studies of dyes and aromatic compounds onto fly ash, *J. Environ. Sci. Heal. A* 34 (1999) 1179-1204.
- [40] G. Limousin, J.-P. Gaudet, L. Charlet, S. Szenknect, V. Barthes, M. Krimissa, Sorption isotherms: a review on physical bases, modeling and measurement, *Appl. Geochem.* 22 (2007) 249-275.
- [41] H. Yuh-Shan, Citation review of Lagergren kinetic rate equation on adsorption reactions, *Scientometrics* 59 (2004) 171-177.
- [42] A.B. Albadarin, C. Mangwandi, A.H. Al-Muhtaseb, G.M. Walker, S.J. Allen, M.N.M. Ahmad, Kinetic and thermodynamics of chromium ions adsorption onto low-cost dolomite adsorbent, *Chem. Eng. J.* 179 (2012) 193-202.
- [43] M. Nadeem, A. Mahmood, S.A. Shahid, S.S. Shah, A.M. Khalid, G. McKay, Sorption of lead from aqueous solution by chemically modified carbon adsorbents, *J. Hazard. Mater.* 138 (2006) 604-613.
- [44] C. Aharoni, F.C. Tompkins, Kinetics of adsorption and desorption and the Elovich equation in advances in catalysis, Eds. D.D. Eley, B.W. Paul. Academic Press, 1970, pp. 1-49.
- [45] W. Clayton, S. Chien, Application of Elovich equation to the kinetics of phosphate release and sorption in soils, *Soil Sci. Soc. Am. J.* 44 (1980) 265-268.
- [46] H. Teng, C.T. Hsieh, Activation energy for oxygen chemisorption on carbon at low temperatures, *Ind. Eng. Chem. Res.* 38 (1998) 292-297.
- [47] G.J. Millar, A. Winnett, T. Thompson, S.J. Couperthwaite, Equilibrium studies of ammonium exchange with Australian natural zeolites, *J. Water Process Eng.* 9 (2016) 47-57.
- [48] J. Lladó, C. Lao-Luque, B. Ruiz, E. Fuente, M. Solé-Sardans, A.D. Dorado, Role of activated carbon properties in atrazine and paracetamol adsorption equilibrium and kinetics, *Process Saf. Environ.* 95 (2015) 51-59.

- [49] L. Al-Khateeb, A.Y. Obaid, N.A. Asiri, M.A. Salam, Adsorption behaviour of estrogenic compounds on carbon nanotubes from aqueous solutions: Kinetic and thermodynamic studies, *J. Ind. Eng. Chem.* 20 (2014) 916-924.
- [50] L. Chen, K.-S. Zhang, J.-Y. He, W.-H. Xu, X.-J. Huang, J.-H. Liu, Enhanced fluoride removal from water by sulfate-doped hydroxyapatite hierarchical hollow microspheres, *Chem. Eng. J.* 285 (2016) 616-624.
- [51] H.-T. Fan, W. Sun, B. Jiang, Q.-J. Wang, D.-W. Li, C.-C. Huang, K.-J. Wang, Z.-G. Zhang, W.-X. Li, Adsorption of antimony (III) from aqueous solution by mercapto-functionalized silica-supported organic-inorganic hybrid sorbent: Mechanism insights, *Chem. Eng. J.* 286 (2016) 128-138.
- [52] A. Chatterjee, S. Schiewer, Multi-resistance kinetic models for biosorption of Cd by raw and immobilized citrus peels in batch and packed-bed columns, *Chem. Eng. J.* 244 (2014) 105-116.
- [53] C. Yao, T. Chen, A new simplified method for estimating film mass transfer and surface diffusion coefficients from batch and adsorption kinetic data, *Chem. Eng. J.* 265 (2015) 93-99.
- [54] M. Rathod, S. Halder, S. Basha, Nanocrystalline cellulose for removal of tetracycline hydrochloride from water via biosorption: Equilibrium, kinetic and thermodynamic studies, *Ecol. Eng.* 84 (2015) 240-249.
- [55] S. Álvarez-Torrellas, A. Rodríguez, G. Ovejero, J. García, Comparative adsorption performance of ibuprofen and tetracycline from aqueous solution by carbonaceous materials, *Chem. Eng. J.* 283 (2016) 936-947.
- [56] Y. Zhao, X. Gu, S. Li, R. Han, G. Wang, Insights into tetracycline adsorption onto kaolinite and montmorillonite: experiments and modeling, *Environ. Sci. Pollut. Res.* 22 (2015) 17031-17040.
- [57] C.H. Giles, T.H. MacEwan, S.N. Nakhwa, D. Smith, Studies in Adsorption. Part XI: A system of classification of solution adsorption isotherms, and its use in diagnosis of adsorption mechanisms and in measurement of specific surface areas of solids", *J. Chem. Soc.* 60 (1960) 3973-3993.
- [58] V. Gómez, M.S. Larrechi, M.P. Callao, Kinetic and adsorption study of acid dye removal using activated carbon, *Chemosphere* 69 (2007) 1151-1158.
- [59] L.N. Konda, I. Czinkota, G. Fülek, G. Morovján, Modeling of single-step and multistep adsorption isotherms of organic pesticides on soil, *J. Agr. Food Chem.* 50 (2002) 7326-7331.

- [60] R. Sips, On the structure of a catalyst surface, J. Chem. Phys. 16 (1948) 490-495.
- [61] S. Tangsir, L.D. Hafshejani, A. Lähde, M. Maljanen, A. Hooshmand, A.A. Naseri, H. Moazed, J. Jokiniemi, A. Bhatnagar, Water defluoridation using Al_2O_3 nanoparticles synthesized by flame spray pyrolysis (FSP) method, Chem. Eng. J. 288 (2016) 198-206.

ACCEPTED MANUSCRIPT

Table 1. Physico-chemical properties of flumequine and tetracycline.

Compound	Structure	Use/ category	Molecular weight (g.mol ⁻¹)	Log K _{ow}	pKa	Molecular size (nm)
Flumequine		Antibiotic	261.25	1.7 ^a	6.4 ^b	1.0x1.0x0.38
Tetracycline		Antibiotic	444.43	-1.30 ^c	3.30, 7.68, 9.68 ^d	1.29x0.62x0.75 ^e

^a [25] Tolls, 2001.^b [26] Chiou, 1989.^c [27] Ueda et al., 1985.^d [28] Li et al., 2013.^e [29] Gambinossi et al., 2004.

Table 2. Specific surface area (S_{BET}), non-microporous surface area ($S_{\text{Non-mic}}$), micropore volume (V_{Mic}), total pore volume (V_{Total}) and pore size obtained from pore size distributions of the tested materials.

Material	S_{BET} ($\text{m}^2 \cdot \text{g}^{-1}$)	$S_{\text{Non-mic}}$ ($\text{m}^2 \cdot \text{g}^{-1}$)	V_{Mic} ($\text{cm}^3 \cdot \text{g}^{-1}$)	V_{Total} ($\text{cm}^3 \cdot \text{g}^{-1}$)	$V_{\text{Mic}}/V_{\text{Total}}$	Average pore width (nm)
GBCM	10	-	-	0.02	-	-
GBCM₂₀₀	352	24	0.170	0.1704	0.997	0.86
GBCM₃₀₀	391	65	0.192	0.1933	0.994	1.10
GBCM₃₅₀	436	123	0.217	0.2354	0.921	1.31

Table 3. Raman spectroscopy, elemental analysis and pH_{IEP} data of the tested materials.

Material	Raman spectroscopy	Elemental analysis					pH_{IEP}
	R	N (%)	C (%)	H (%)	O (%)	S (%)	
GBCM ₂₀₀	1.83	0.78	84.23	1.05	12.54	1.40	5.6
GBCM ₃₀₀	1.93	0.50	79.97	0.90	17.52	1.11	3.8
GBCM ₃₅₀	2.25	0.33	78.75	0.70	19.32	0.90	2.6

Table 4. Kinetic model parameters (linearized form equations) for the adsorption of flumequine and tetracycline onto GBCM₃₀₀.

Fractional power model						
	a (mg.g ⁻¹ .h ⁻¹)		b		R ²	
Flumequine	3.23		0.43		0.8987	
Tetracycline	4.32		0.31		0.9790	
Lagergren pseudo-first order model						
	q _e exp (mg.g ⁻¹)	q _e cal (mg.g ⁻¹)	k ₁ (h ⁻¹)		R ²	
Flumequine	17.80	14.34	0.06		0.8266	
Tetracycline	20.25	20.59	0.04		0.8181	
Pseudo-second order model						
	q _e exp (mg.g ⁻¹)	q _e cal (mg.g ⁻¹)	k ₂ (g.mg ⁻¹ .h ⁻¹)	t _{1/2} (h)	v ₀ (mg.g ⁻¹ .h ⁻¹)	R ²
Flumequine	17.80	19.61	0.006	8.10	2.00	0.9834
Tetracycline	20.25	21.69	0.003	14.87	1.27	0.9520
Elovich model						
	α (g.mg ⁻¹ .h ⁻¹)		β (mg.g ⁻¹ .h ⁻¹)		R ²	
Flumequine	0.78		3.39		0.9754	
Tetracycline	1.00		3.03		0.8808	

Table 5. Kinetic model parameters (non-linearized form equations) for the adsorption of flumequine and tetracycline onto GBCM₃₀₀.

Fractional power model						
	q _e exp (mg.g ⁻¹)	q _e cal (mg.g ⁻¹)	a (mg.g ⁻¹ .h ⁻¹)	b	SSE	
Flumequine	17.80	19.79	4.68	0.32	26.97	
Tetracycline	20.25	20.22	3.71	0.35	7.98	
Lagergren pseudo-first order model						
	q _e exp (mg.g ⁻¹)	q _e cal (mg.g ⁻¹)	k ₁ (h ⁻¹)		SSE	
Flumequine	17.80	17.80	0.14		17.43	
Tetracycline	20.25	19.85	0.03		70.66	
Pseudo-second order model						
	q _e exp (mg.g ⁻¹)	q _e cal (mg.g ⁻¹)	k ₂ (g.mg ⁻¹ .h ⁻¹)	t _{1/2} (h)	v ₀ (mg.g ⁻¹ .h ⁻¹)	SSE
Flumequine	17.80	16.94	0.012	5.08	3.68	8.91
Tetracycline	20.25	18.13	0.004	15.68	1.44	54.37
Elovich model						
	q _e exp (mg.g ⁻¹)	q _e cal (mg.g ⁻¹)	α (g.mg ⁻¹ .h ⁻¹)	β (mg.g ⁻¹ .h ⁻¹)		SSE
Flumequine	17.80	18.77	0.78	3.39		8.92
Tetracycline	20.25	17.88	1.00	3.03		40.61

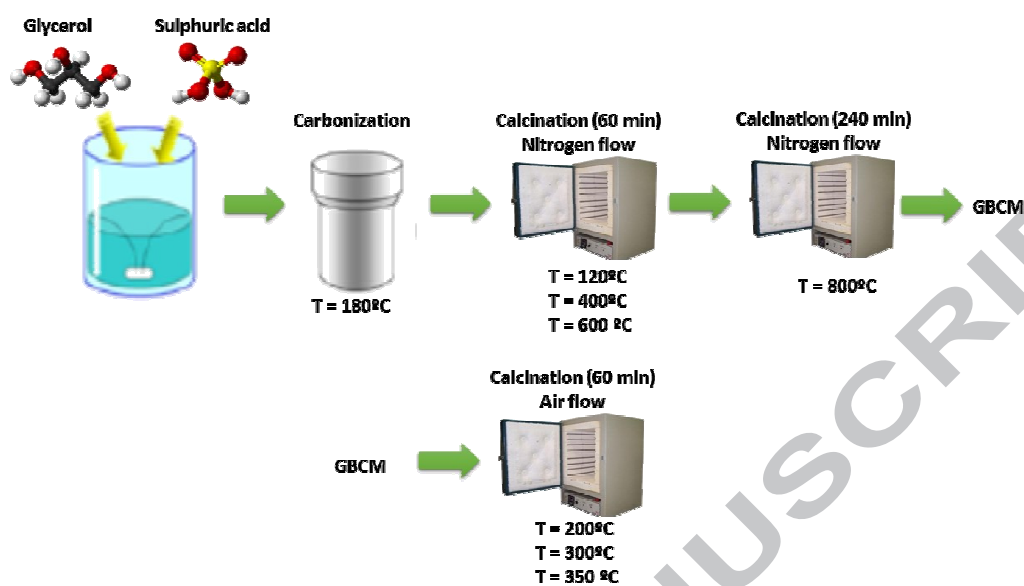
Table 6. Parameters of intra-particle diffusion model and estimated mass transfer coefficients for the adsorption of flumequine and tetracycline onto GBCM₃₀₀ material.

Compound	$k_{i,1}$ (mg.g ⁻¹ .h ^{-0.5})	R^2_1	$k_{i,2}$ (mg.g ⁻¹ .h ^{-0.5})	R^2_2
Flumequine	4.31	0.9556	1.61	0.9638
Tetracycline	2.90	0.9113	1.43	0.9756
Compound	r_0 (mg.g ⁻¹ .h ⁻¹)	k_e (h ⁻¹)	$k_f \times 10^{-6}$ (m.s ⁻¹)	$D_s \times 10^{-14}$ (m ² .s ⁻¹)
Flumequine	3.01	0.06	1.09	3.88
Tetracycline	5.41	0.08	1.97	5.06

Table 7. Adsorption isotherms parameters for the adsorption of flumequine and tetracycline onto the glycerol-based carbon materials.

Flumequine										
Material	q_{sat} (mg.g⁻¹)	b (L.mg⁻¹)	SSE	K_F (L.g⁻¹)	n_F	SSE	q_m (mg.g⁻¹)	K_S (L.mg⁻¹)	n	SSE
GBCM₂₀₀	0.71	0.57	0.1	0.44	8.71	0.2	0.69	0.37	0.64	0.1
GBCM₃₀₀	-	-	-	-	-	-	481.98	0.01	0.17	105.0
GBCM₃₅₀	2.60 x 10 ⁵	9.24 x 10 ⁻⁷	493.3	0.001	0.44	337.0	7,439	9 x 10 ⁻⁴	0.44	337.2
Tetracycline										
Material	q_{sat} (mg.g⁻¹)	b (L.mg⁻¹)	SSE	K_F (L.g⁻¹)	n_F	SSE	q_m (mg.g⁻¹)	K_S (L.mg⁻¹)	n	SSE
GBCM₂₀₀	-	-	-	-	-	-	31.69	0.06	1.07	13.4
GBCM₃₀₀	-	-	-	7.74 x 10 ⁻⁹	0.21	750.6	9,754	0.008	0.04	373.4
GBCM₃₅₀	-	-	-	0.15	0.77	418.8	6,797	0.28 x 10 ⁻³	0.76	419.7

Figure captions**Scheme 1.** Synthesis of GBCM.**Fig 1.** N₂ adsorption-desorption isotherms at -196 °C of the tested glycerol-based carbon materials.**Fig 2.** SEM micrographs of (a) GBCM₃₀₀ and (b) GBCM₃₅₀.**Fig 3.** FT-IR spectra of the tested glycerol-based carbon materials.**Fig. 4.** Raman spectra of the tested glycerol-based carbon materials.**Fig 5.** Kinetic experimental and theoretical adsorption data (linearized form equations) for the removal of (a) flumequine and (b) tetracycline onto GBCM₃₀₀ carbon. (Conditions: initial concentration = 100 mg.L⁻¹; pH = 6.3; particle size = 425-600 µm; adsorbent dosage = 2.4 g.L⁻¹; temperature= 30 °C; stirring rate = 300 rpm).**Fig. 6.** Kinetic experimental and theoretical adsorption data (non-linearized form equations) for the removal of (a) flumequine (FLQ) and (b) tetracycline (TCN) onto GBCM₃₀₀ carbon (Conditions: initial concentration = 100 mg.L⁻¹; pH = 6.3; particle size = 425-600 µm; adsorbent dosage = 2.4 g.L⁻¹; temperature = 30 °C; stirring rate = 300 rpm).**Fig. 7.** FTIR spectra of GBCM₃₀₀ carbon material before and after adsorption of flumequine (FLQ) and tetracycline (TCN).**Fig. 8.** Intra-particle diffusion model (Weber and Morris plot) for the adsorption of flumequine and tetracycline onto GBCM₃₀₀.**Fig. 9.** Relation between the adsorption capacity of flumequine and tetracycline and the specific surface area of GBCM₂₀₀, GBCM₃₀₀ and GBCM₃₅₀ carbon materials.**Fig. 10.** Experimental and theoretical adsorption isotherms of (a) flumequine (FLQ) and (b) tetracycline (TCN) onto the glycerol-based carbon materials (Conditions: initial concentration = 100 mg.L⁻¹; pH = 6.3; particle size = 425-600 µm; temperature = 30 °C; stirring rate = 300 rpm).**Fig. 11.** Comparison of adsorption capacities of flumequine (FLQ) and tetracycline (TCN) onto the glycerol-based carbon materials.



Scheme 1. Synthesis of GBCM.

a)

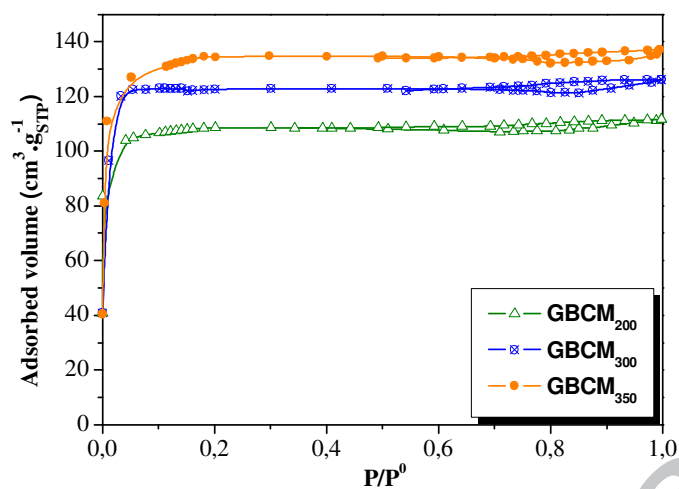
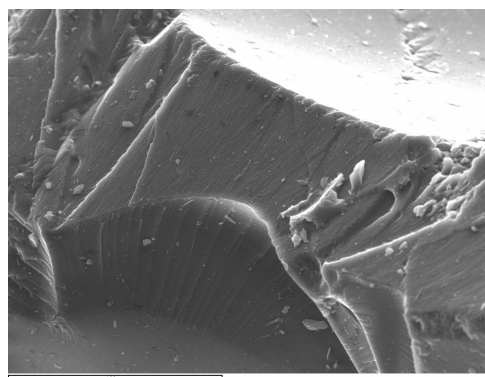
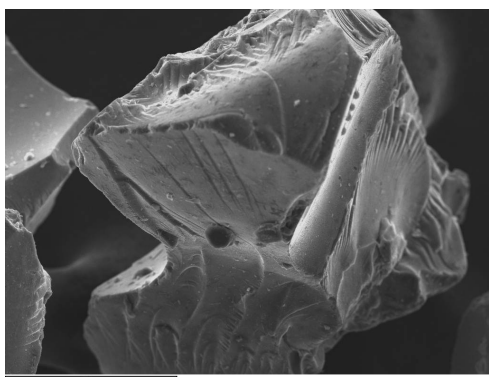


Fig. 1. N₂ adsorption-desorption isotherms at -196 °C of the tested glycerol-based carbon materials.

a)



b)

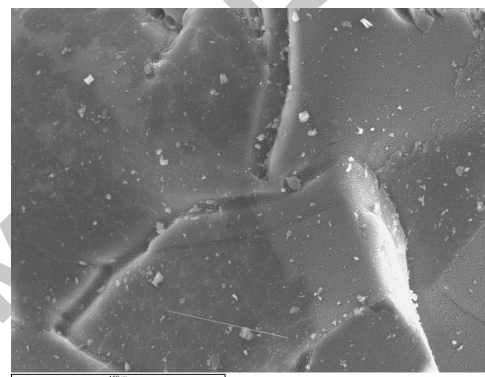
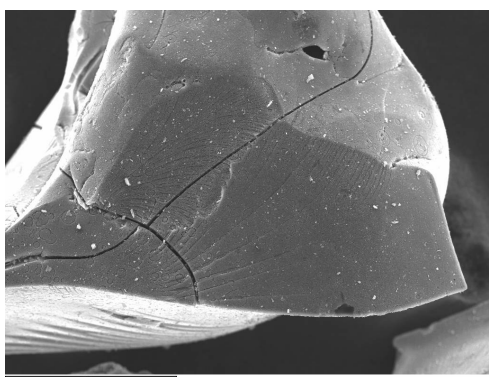


Fig. 2. SEM micrographs of (a) GBCM₃₀₀ and (b) GBCM₃₅₀.

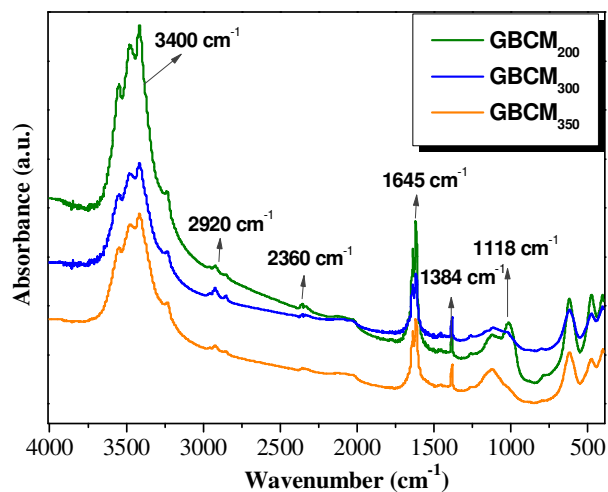


Fig. 3. FT-IR spectra of the tested glycerol-based carbon materials.

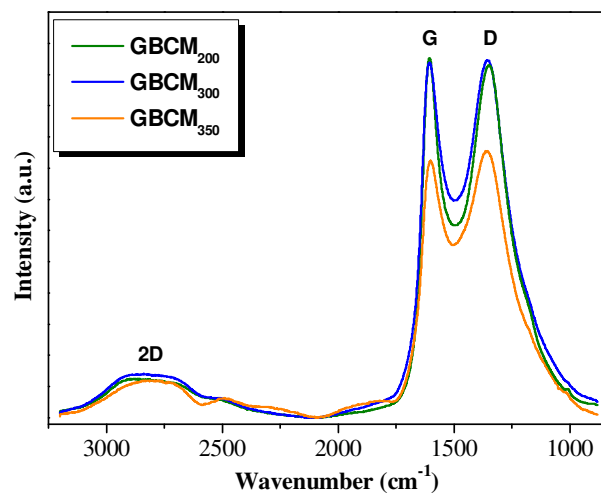


Fig. 4. Raman spectra of the tested glycerol-based carbon materials.

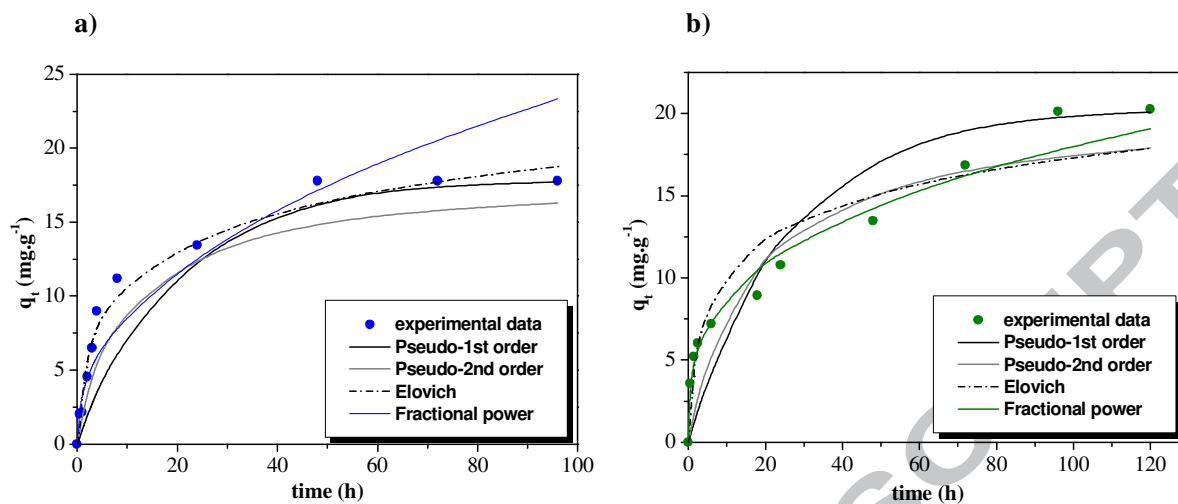


Fig. 5. Kinetic experimental and theoretical adsorption data (linearized form equations) for the removal of (a) flumequine (FLQ) and (b) tetracycline (TCN) onto GBCM₃₀₀ carbon (Conditions: initial concentration = 100 mg.L⁻¹; pH = 6.3; particle size = 425-600 μ m; adsorbent dosage = 2.4 g.L⁻¹; temperature = 30 °C; stirring rate = 300 rpm).

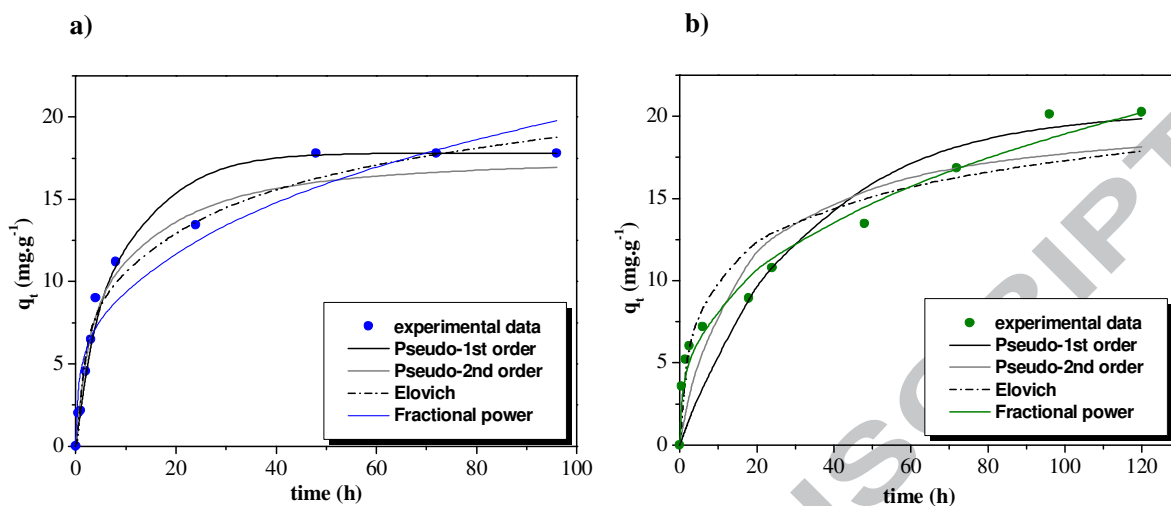


Fig. 6. Kinetic experimental and theoretical adsorption data (non-linearized form equations) for the removal of (a) flumequine (FLQ) and (b) tetracycline (TCN) onto GBCM₃₀₀ carbon (Conditions: initial concentration = 100 mg.L⁻¹; pH = 6.3; particle size = 425-600 μ m; adsorbent dosage = 2.4 g.L⁻¹; temperature = 30 °C; stirring rate = 300 rpm).

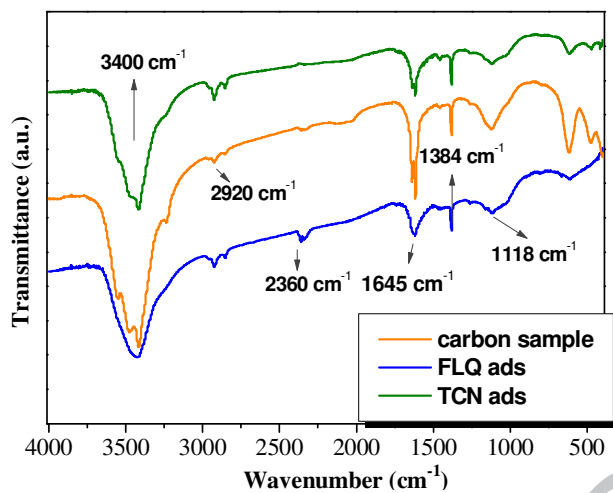


Fig. 7. FTIR spectra of GBCM₃₀₀ carbon material before and after adsorption of flumequine (FLQ) and tetracycline (TCN).

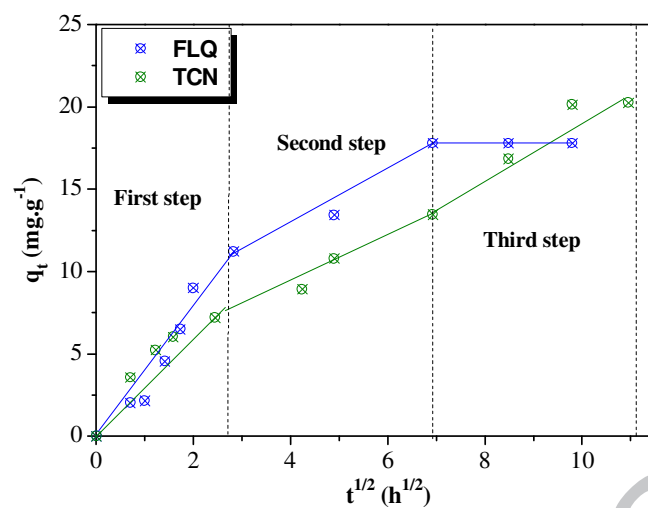


Fig. 8. Intra-particle diffusion model (Weber and Morris plot) for the adsorption of flumequine and tetracycline onto GBCM₃₀₀.

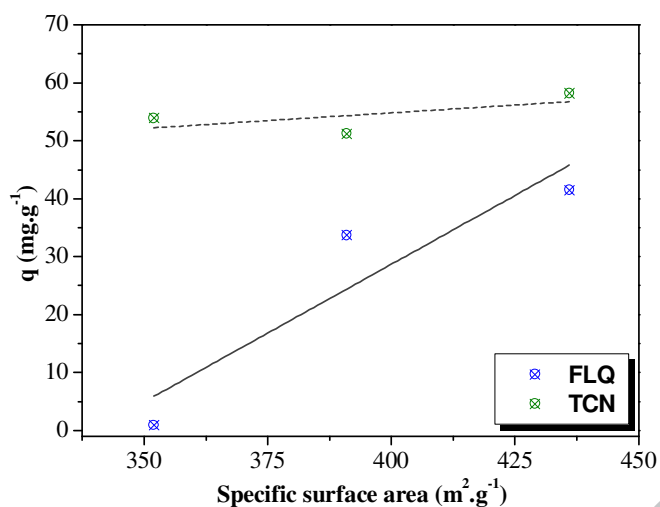


Fig. 9. Relation between the adsorption capacity of flumequine and tetracycline and the specific surface area of GBCM_{200} , GBCM_{300} and GBCM_{350} carbon materials.

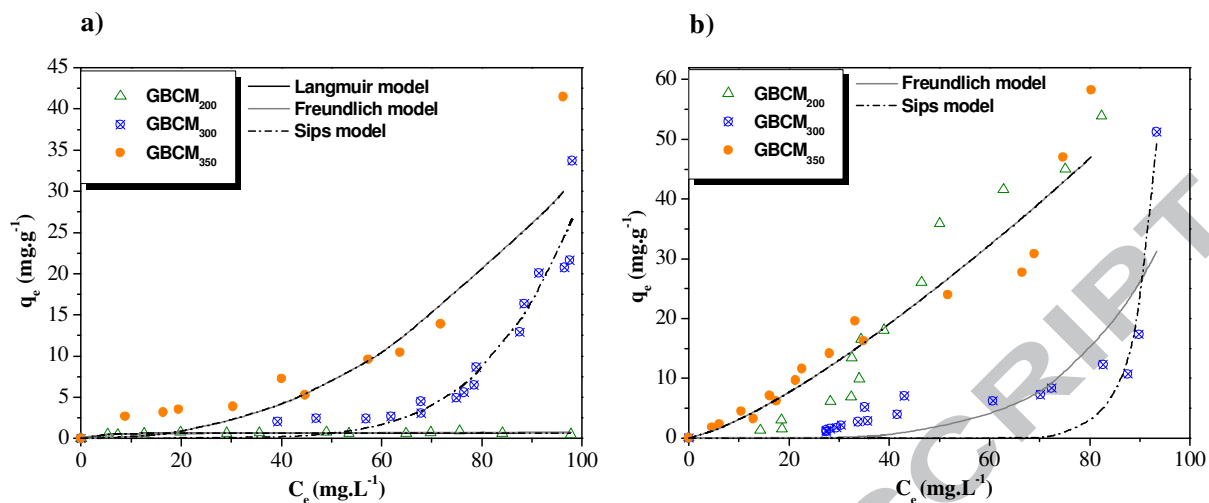


Fig. 10. Experimental and theoretical adsorption isotherms of (a) flumequine (FLQ) and (b) tetracycline (TCN) onto the glycerol-based carbon materials (Conditions: initial concentration = 100 mg.L⁻¹; pH = 6.3; particle size = 425-600 μ m; temperature = 30 $^{\circ}$ C; stirring rate = 300 rpm).

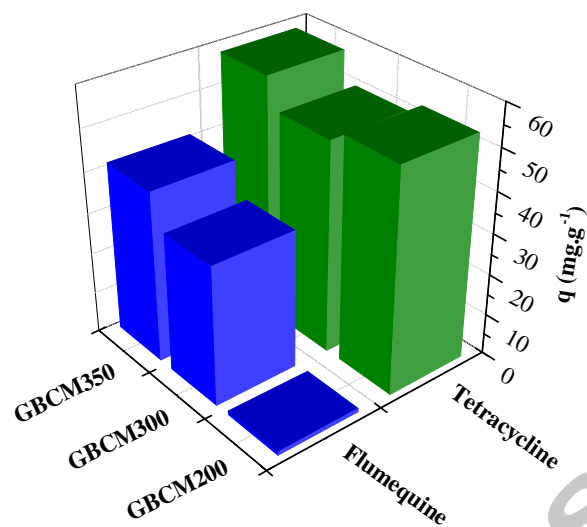
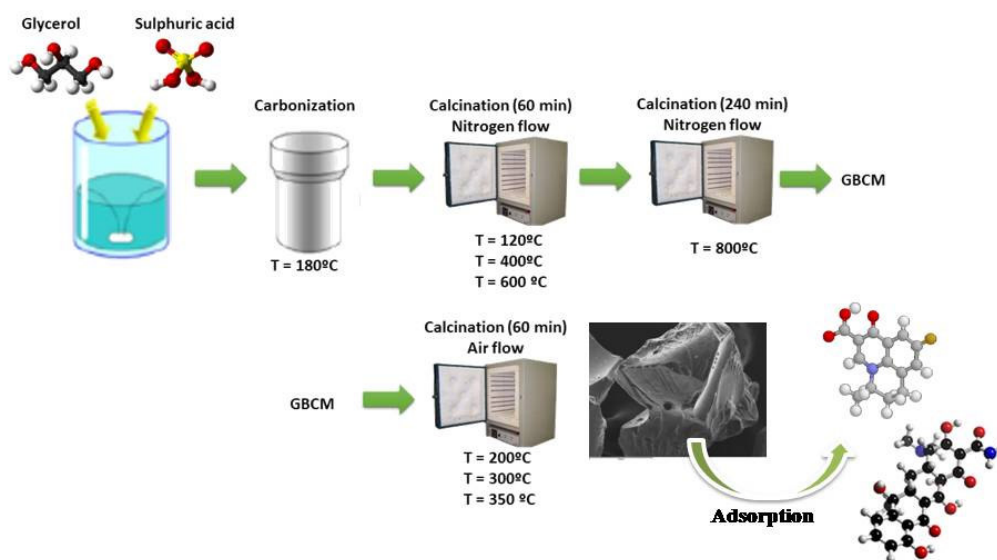


Fig. 11. Comparison of adsorption capacities of flumequine (FLQ) and tetracycline (TCN) onto the glycerol-based carbon materials.

Graphical abstract



Highlights

First work studying the removal of antibiotics by carbonaceous materials prepared from glycerol.> Kinetic and equilibrium adsorption tests onto glycerol-based carbon materials.> Textural and chemical adsorbents properties played a critical role in the adsorption performance.> Film mass transfer and pore diffusion coefficients were estimated.

ACCEPTED MANUSCRIPT